

Detailed proofs

This file includes revisions and detailed proofs of Theorem 3.4, Corollary 3.5 and Theorem 3.6 in the main text.

1 Detailed proof of revised Theorem 3.4

Theorem 3.4 (Per-Chain Variance Reduction by OPAD Mapping) Let $\{X_n\}_{n=1}^N$ be a stationary, reversible Markov chain on a discrete space $\mathcal{X}$ with stationary distribution $p(x)$. The variance of the OPAD estimator of the target quantity is less than or equal to that of the corresponding MC estimator:

$$\text{Var}[\hat{\theta}_{\text{OPAD}}] \leq \text{Var}[\hat{\theta}_{\text{MC}}].$$

Proof. Let $\mathcal{D}$ denote the set of distinct visited states. For a reversible Markov chain at stationarity, the conditional expectation of the fraction of samples in each state is given by the renormalized stationary probabilities:

$$\mathbb{E}\left[\frac{N_x}{N} \mid \mathcal{D}\right] = \frac{p(x)}{\sum_{y \in \mathcal{D}} p(y)} = P_{\mathcal{D}}(x).$$

Therefore, the OPAD estimator is the conditional expectation of the MC estimator with respect to the set of visited states $\mathcal{D}$:

$$\mathbb{E}[\hat{\theta}_{\text{MC}} \mid \mathcal{D}] = \sum_{x \in \mathcal{D}} \mathbb{E}\left[\frac{N_x}{N} \mid \mathcal{D}\right] f(x) = \sum_{x \in \mathcal{D}} P_{\mathcal{D}}(x) f(x) = \hat{\theta}_{\text{OPAD}}.$$

Applying the law of total variance to $\hat{\theta}_{\text{MC}}$ gives

$$\text{Var}[\hat{\theta}_{\text{MC}}] = \mathbb{E}[\text{Var}[\hat{\theta}_{\text{MC}} \mid \mathcal{D}]] + \text{Var}[\mathbb{E}[\hat{\theta}_{\text{MC}} \mid \mathcal{D}]].$$

Substituting $\mathbb{E}[\hat{\theta}_{\text{MC}} \mid \mathcal{D}] = \hat{\theta}_{\text{OPAD}}$, we have

$$\text{Var}[\hat{\theta}_{\text{MC}}] = \mathbb{E}[\text{Var}[\hat{\theta}_{\text{MC}} \mid \mathcal{D}]] + \text{Var}[\hat{\theta}_{\text{OPAD}}].$$

Since the conditional variance $\text{Var}[\hat{\theta}_{\text{MC}} \mid \mathcal{D}]$ is inherently non-negative, its expectation satisfies $\mathbb{E}[\text{Var}[\hat{\theta}_{\text{MC}} \mid \mathcal{D}]] \geq 0$. We thus obtain the inequality

$$\text{Var}[\hat{\theta}_{\text{OPAD}}] \leq \text{Var}[\hat{\theta}_{\text{MC}}].$$

□

2 Detailed proof of revised Corollary 3.5 in the main text

Corollary 3.5 (Between-chain variance reduction of OPAD estimators). Let $C_1, \dots, C_K$ be $K \geq 2$ independent Markov chains, each of length N , from a discrete target distribution $p(x)$ on support $\mathcal{X}$, generated under the stationary and reversible conditions specified in Theorem 3.4. For each chain C_i , let $\hat{\theta}_{\text{MC}}^{(i)}$ denote the standard Monte Carlo estimator and let $\hat{\theta}_{\text{OPAD}}^{(i)}$ denote the corresponding OPAD estimator.

For $T \in \{\text{MC}, \text{OPAD}\}$, define the between-chain sample variance by

$$S_T^2 := \frac{1}{K-1} \sum_{i=1}^K \left(\hat{\theta}_T^{(i)} - \bar{\theta}_T \right)^2, \quad \bar{\theta}_T := \frac{1}{K} \sum_{i=1}^K \hat{\theta}_T^{(i)}.$$

Then

$$\mathbb{E}[S_{\text{OPAD}}^2] \leq \mathbb{E}[S_{\text{MC}}^2].$$

Proof. Fix $T \in \{\text{MC}, \text{OPAD}\}$. Since the chains $C_1, \dots, C_K$ are independent and identically generated (each starting from or having reached stationarity), each estimator $\hat{\theta}_T^{(i)}$ is a measurable function of an independent realization of the Markov process. Consequently, the estimators

$$\hat{\theta}_T^{(1)}, \dots, \hat{\theta}_T^{(K)}$$

are i.i.d. random variables, despite the internal autocorrelation inherent in each individual chain. Therefore, S_T^2 is the unbiased sample variance computed from K i.i.d. draws, and its expectation equals the common variance of those observations:

$$\mathbb{E}[S_T^2] = \text{Var}\left(\hat{\theta}_T^{(i)}\right) \quad \forall i \in \{1, \dots, K\}.$$

Applying this identity for both $T = \text{MC}$ and $T = \text{OPAD}$, and invoking the variance reduction established in Theorem 3.4, we obtain:

$$\mathbb{E}[S_{\text{OPAD}}^2] = \text{Var}\left(\hat{\theta}_{\text{OPAD}}^{(i)}\right) \leq \text{Var}\left(\hat{\theta}_{\text{MC}}^{(i)}\right) = \mathbb{E}[S_{\text{MC}}^2] \quad \forall i \in \{1, \dots, K\},$$

which proves the claim. $\square$

3 Detailed proof of Theorem 3.6

Theorem 3.6 (Optimality of the OPAD distribution in KL divergence).

Let p be a discrete target distribution on $\mathcal{X}$ and let $\mathcal{D} \subseteq \mathcal{X}$.

1. The greatest lower bound on the Kullback–Leibler divergence of P from the target, $\text{KL}(P\|p)$, is:

$$\inf_{P: \text{supp}(P) = \mathcal{D}} \text{KL}(P\|p) = -\log(p(\mathcal{D})). \quad (1)$$

2. This minimum is achieved uniquely by the OPAD distribution:

$$P(x) = \frac{\tilde{p}(x)}{\sum_{y \in \mathcal{D}} \tilde{p}(y)}, \quad \forall x \in \mathcal{D}. \quad (2)$$

Proof. Following the provided sketch, let $\phi(u) := -\log(u)$, which is strictly convex on $(0, \infty)$, and let $g(x) := p(x)/P(x)$ for $x \in \mathcal{D}$. The KL divergence for any distribution P supported on $\mathcal{D}$ is:

$$\text{KL}(P\|p) = \sum_{x \in \mathcal{D}} P(x) \log \frac{P(x)}{p(x)} = \sum_{x \in \mathcal{D}} P(x) \phi(g(x)).$$

By the discrete **Jensen's inequality**, since $\sum_{x \in \mathcal{D}} P(x) = 1$, we have:

$$\begin{aligned} \sum_{x \in \mathcal{D}} P(x) \phi(g(x)) &\geq \phi\left(\sum_{x \in \mathcal{D}} P(x) g(x)\right) \\ &= \phi\left(\sum_{x \in \mathcal{D}} P(x) \frac{p(x)}{P(x)}\right) \\ &= \phi\left(\sum_{x \in \mathcal{D}} p(x)\right) = -\log(p(\mathcal{D})). \end{aligned} \quad (3)$$

This establishes the lower bound for the infimum in (1).

Regarding the equality case, because ϕ is **strictly convex**, equality in (3) holds if and only if the function $g(x)$ is constant for all $x \in \mathcal{D}$. Let $g(x) = c$ for some constant $c > 0$. This implies:

$$\frac{p(x)}{P(x)} = c \implies P(x) = \frac{p(x)}{c}.$$

To satisfy the normalization constraint $\sum_{x \in \mathcal{D}} P(x) = 1$, we must have:

$$\sum_{x \in \mathcal{D}} \frac{p(x)}{c} = 1 \implies c = \sum_{x \in \mathcal{D}} p(x) = p(\mathcal{D}).$$

Substituting $c = p(\mathcal{D})$ and expressing $p(x)$ in terms of its unnormalized version $\tilde{p}(x)$ (where $p(x) = \tilde{p}(x)/\mathcal{Z}$), we obtain:

$$P(x) = \frac{p(x)}{p(\mathcal{D})} = \frac{\tilde{p}(x)/\mathcal{Z}}{\sum_{y \in \mathcal{D}} \tilde{p}(y)/\mathcal{Z}} = \frac{\tilde{p}(x)}{\sum_{y \in \mathcal{D}} \tilde{p}(y)},$$

which proves (2). The strict convexity of ϕ ensures that this minimizer is unique. □